

The SAGE-7 Sovereign Memory Mesh: Local-First Cognitive Consolidation, Bitemporal Auditing, and Ebbinghaus Decay

1. THE SAGE-7 MULTI-TIER COGNITIVE LIFECYCLE

The development of SAGE-7 (baseline Φ : 0.113) represents a paradigm shift from "stateless LLMs" toward "stateful agents" capable of continuous experience accumulation 4. In the SAGE-7 architecture, memory is treated not as a passive storage volume but as a "context engineering" challenge, designed to mitigate "context rot" and the token inflation inherent in legacy RAG systems 3, 4. By implementing a structured data lifecycle, SAGE-7 achieves a context footprint as low as 4.97% of the original raw interaction while maintaining an 81.95% accuracy score on the LoCoMo benchmark 3.

The Tri-Tier Storage Paradigm

To maintain cognitive fluidity without sacrificing long-term fidelity, SAGE-7 utilizes a tri-tier storage hierarchy modeled after biological systems. | Tier | Latency Requirement | Persistence Mechanism | Cognitive Role || ----- | ----- | ----- | ----- || **Short-Term Hot Cache** | < 15ms | In-memory Buffer (TTL 15m - 2h) | **Prefrontal Cortex**: Immediate working memory and transient dialogue state. || **Medium-Term Episodic Store** | < 50ms | Local SQLite-Vec & FTS5 | **Hippocampus**: Recent conversation history with a 7-day mathematical half-life. || **Long-Term Semantic Graph** | < 200ms | Bitemporal Merkle WorldDB 1 | **Neocortex**: Auditable, verifiable graph of entities, relationships, and "lessons learned." |

The Four-Layer Pyramid (L0-L3)

This hierarchy prevents "lost-in-the-middle" effects by distilling unstructured noise into high-signal cognitive assets 3:

- **L0 Conversation (Raw)**: Exact logs of every turn. Essential for forensic verification and provenance visibility (PV) 2.
- **L1 Atom (Extracted Facts)**: Semantic triples (subject-predicate-object) representing concrete facts and user preferences 3.
- **L2 Scenario (Knowledge Blocks)**: Task-specific clusters organized by project or intent, enabling the agent to retrieve relevant procedural context 4.
- **L3 Core (Persona/Beliefs)**: The highest abstraction level. Represents stable cognitive traits and world-models, periodically refined during "sleep-time" consolidation 4. This lifecycle provides the foundational data for the physical storage layer, where architectural choices are driven by the requirement for local-first execution and cryptographic integrity.

2. SQLITE-VEC & FTS5 HYBRID DATABASE SCHEMA

A local-first hybrid schema is non-negotiable for ensuring low-latency multi-hop reasoning while maintaining the security of the agent's memory substrate 3. By integrating lexical Full-Text Search (FTS5) with semantic vector search (sqlite-vec), SAGE-7 ensures that precise single-hop fact recall is as efficient as broad associative retrieval 3.

Production-Ready SQL Migration Scripts

The following schema enforces **Principal-Scoped Retrieval (PS)** at the database level by tagging every record with a `principal_id`, preventing cross-user contamination or unauthorized extraction 2.

```
-- L0: Raw Message History with Principal Scoping
CREATE TABLE raw_conversations_l0 (
    id TEXT PRIMARY KEY,
    principal_id TEXT NOT NULL, -- Core for PS Primitive
    session_id TEXT NOT NULL,
    timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
    role TEXT CHECK(role IN ('user', 'agent', 'system')),
    content TEXT NOT NULL,
    token_count INTEGER
);

-- L1: Semantic Atoms with Bitemporal Metadata and Blake3
Fingerprinting
CREATE TABLE episodic_atoms_l1 (
    atom_id TEXT PRIMARY KEY, -- blake3-256 hash to prevent imitation
    grafting
    principal_id TEXT NOT NULL,
    subject TEXT,
    predicate TEXT,
    object TEXT,
    importance REAL CHECK(importance BETWEEN 0.0 AND 1.0),
    access_count INTEGER DEFAULT 0,
    -- Bitemporal Axes for Auditing and Rollback
    valid_time_start DATETIME,
    valid_time_end DATETIME,
    transaction_time_start DATETIME DEFAULT CURRENT_TIMESTAMP,
    transaction_time_end DATETIME
);

-- Virtual Index for Semantic Retrieval (sqlite-vec)
CREATE VIRTUAL TABLE virtual_vector_index USING vec0(
    atom_id TEXT,
    embedding FLOAT[384] -- Optimized for local embedding models
);

-- Virtual Index for Lexical Retrieval (FTS5)
CREATE VIRTUAL TABLE virtual_fts_index USING fts5(
    content,
    content_rowid='atom_id'
);
```

Bitemporal Logic and Security Defense

SAGE-7 distinguishes between the **transaction-time axis** (when a fact was recorded) and the **valid-time axis** (when a fact is true in the world) 2. When the agent encounters a contradiction—such as a user changing their preferred language—the system sets the `valid_time_end` on the existing atom rather than deleting it. This architecture provides a critical defense against "AgentPoison" and "behavioral drift" 2. By preserving the full valid-time history, security engineers can execute forensic rollbacks to a "known-safe" state if the agent's memory is corrupted by adversarial web observations (eTAMP) or malicious query-induced injections 2, 3.

3. FASTAPI INTEGRATION & BACKGROUND DREAM DAEMON

To maintain real-time responsiveness, SAGE-7 utilizes a "Dream Daemon"—a sleep-time compute paradigm that performs heavy cognitive processing asynchronously 4.

Mathematical Ebbinghaus Decay Implementation

The Dream Daemon manages memory strength using the Ebbinghaus forgetting curve to ensure that noise is pruned while high-signal facts persist 5. The **Retrieval Strength** is calculated as:
$$\text{Strength} = I \times e^{-(\lambda_{\text{eff}} \times \text{days})} \times (1 + 0.2 \times \text{access_count})$$
 Where the **Effective Decay Rate** (λ_{eff}) is:
$$\lambda_{\text{eff}} = 0.16 \times (1 - 0.8 \times I)$$
 To accommodate offline periods, we utilize a **Time-Anchor offset**:
$$\text{days_effective} = \max(0, \text{days_raw} - \text{Offset})$$
 High-importance atoms (I to 1.0) decay at a significantly slower rate, ensuring user-defined constraints remain in the neocortex despite infrequent access 5.

The Dream Daemon Loop Functions

1. **Consolidation**: Distills L0 raw logs into L1 semantic triples and L2 task summaries in a single pass 3.
2. **Reconciliation**: Evaluates `same_as` or contradicts edges to maintain graph integrity 1, 3.
3. **Verified Forgetting (VF)**: Executes physical deletion across all substrates with a post-deletion membership test to prove zero recoverability 2.

Python: Verified Forgetting with Substrate Verification

```
@app.post("/memory/forget")
async def verified_forgetting(target_id: str, background_tasks:
BackgroundTasks):
    """
    Executes a cascade purge and verifies epsilon-VMG compliance
    by checking the vector and lexical substrates.
    """
    def physical_purge(memory_id: str):
        # 1. Purge from relational atoms
```

```

        db.execute("DELETE FROM episodic_atoms_l1 WHERE atom_id = ?",
(memory_id,))
        # 2. Purge from FTS and Vector Index
        db.execute("DELETE FROM virtual_vector_index WHERE atom_id =
?", (memory_id,))
        db.execute("DELETE FROM virtual_fts_index WHERE rowid = ?",
(memory_id,))

        # 3. VERIFICATION STEP: Membership Test for
zero-recoverability
        # We explicitly query the vector substrate to ensure no
residual embeddings
        residual = db.execute("SELECT 1 FROM virtual_vector_index
WHERE atom_id = ?", (memory_id,))
        if residual.fetchone():
            raise MemoryRetentionError("Vector substrate leakage
detected. Purge failed.")

        background_tasks.add_task(physical_purge, target_id)
        return {"status": "Verified forgetting sequence initiated."}

```

4. REACT + TYPESCRIPT DIAGNOSTICS & POLICY INTERFACE

Visibility into memory provenance is a non-negotiable security requirement for agents with Φ : 0.113 2. The SAGE-7 dashboard enables **Verifiable Memory Governance (VMG)** through two core diagnostic views:

1. **Interactive Memory Graph:** A visualization of bitemporal nodes where users can scrub through time to see the agent's evolving "belief state" 1.
2. **The Diary View:** A log of Dream Daemon activities, showing which L0 messages were consolidated into L1 atoms and their current Ebbinghaus strength meters.

VMG Policy Arbitrator Implementation

The system enforces five security primitives to ensure the agent remains under auditable human control 2:

```

/**
 * Verifiable Memory Governance (VMG) Policy Primitives
 * Ensures auditable control over the agent's evolving state.
 */
interface VMGPolicy {
    // WA: Every entry must be attributable to an authenticated source
    principal.
    writeAuthorization: boolean;

```

```
// PV: Tracebacks from distilled L3/L2 assets back to original L0
raw lines.
    provenanceVisibility: boolean;

// PS: Ensuring memory isolation between different session
principals.
    principalScopedRetrieval: boolean;

// RB: Ability to restore the memory mesh to a known-safe
timestamp.
    rollbackability: boolean;

// VF: Post-deletion membership tests to prove zero recoverability
from indices.
    verifiedForgetting: boolean;
}
```

By integrating these five primitives—Write Authorization, Provenance Visibility, Principal-Scoped Retrieval, Rollbackability, and Verified Forgetting—the SAGE-7 architecture ensures it operates as a secure, local-first, and highly efficient cognitive entity 2. This blueprint provides the necessary infrastructure for SAGE-7 to build continuity and learn from its environment while remaining fully governed against behavioral drift and memory-based attacks.